CURRICULUM VITAE

Douglas Gray

Principal Applied Scientist

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Portola Valley, California | Google Scholar | LinkedIn

Applied scientist and research leader working across multimodal retrieval, vision-language systems, generative models, and real-time interfaces. Led science organizations of up to 20 people and built the systems behind Amazon Lens, StyleSnap, More Like This, and visual search suggestions. Experience spans product-scale deep learning, mobile computer vision, relevance evaluation, technical architecture, and development of senior scientific talent. Research interests include multimodal reasoning, agentic retrieval, world models, and persistent visual assistants.

Professional & Research Experience

Principal Applied Scientist 2025-June 2026

Amazon Visual Search

- Originated Amazon's real-time AI image generation in search autocomplete, turning descriptive shopping queries into tappable visual suggestions for similar products; launched to select customers 124 days after goal setting.
- Defined the high-level architecture, GPU-saving prefix cache, safety mechanisms, and quality evaluation; coordinated a 45-person organization across seven parallel workstreams to deliver the launch.
- Served as principal technical lead for Lens Live, a continuous visual search experience integrated with Rufus for product insights; launched to tens of millions of customers.
- Architected the mobile ML pipeline for on-device product detection, tracking, and session state management, enabling continuous low-latency identification as the camera moved.
- Established technical ownership across Lens and Rufus teams in Palo Alto, Seattle, New York, and London: product identification and recognition in Lens, with LLM-generated product insights and follow-up interactions in Rufus.
- Co-authored work on universal generative multimodal retrieval (CVPR 2025) and reinforcement learning for self-verification in multimodal reasoning agents (ECCV 2026).

Senior Manager, Applied Science 2020-2025

Amazon Visual Shopping

- Led an organization of 15-20 applied scientists and engineers, including an applied science manager and up to four software engineers; owned science goals, research direction, and technical artifacts for visual search.
- Completed the transformation of Amazon Lens into a unified deep-learning, embedding-based retrieval system over more than 1 billion products, retiring the classical computer vision pipeline.
- Led an approximately 18-person results-quality program in 2023 that increased results-page click-through rate from roughly 19% to about 40% during the goal period and improved retention by four percentage points; Lens subsequently exceeded 1 billion annual queries.

- Built the relevance plan across embedding training data, a linear-to-GBDT ranker upgrade, category-aware interfaces, barcode results, and evaluation; combined online A/B tests with calibrated human annotation and retrieval metrics.
- Launched multimodal image-plus-text search, Visual Search Suggestions, and More Like This; scaled More Like This from an initial dresses experiment to more than 90% of search-result-page impressions.
- Led the organization responsible for Visual Search Suggestions as it scaled to more than $600 million in incremental revenue measured through A/B tests.
- Developed scientists and engineers across levels, including promotion of a principal engineer in 2022 and a principal scientist in 2024; established written science roadmaps, weekly research reviews, and technical deep dives.
- Connected production systems with external research, including KDD system papers on Shop the Look (2022) and multimodal visual search (2024).

Manager, Applied Science
2016-2020

Amazon Visual Search & AR

- Led up to 12 applied scientists and engineers and introduced deep learning to Amazon's visual search stack, building a new system beyond the flat and printed objects supported by the classical computer vision pipeline.
- Expanded recognition from an initial launch covering approximately 100 product categories to more than 100,000 fine-grained search strings using models trained on Amazon query logs.
- Developed instance-level visual retrieval over approximately 10 million products in apparel and home; launched StyleSnap for fashion in 2019 and StyleSnap for Home in 2020.
- Established a weekly science review built around written experimental updates, technical decisions, and project ownership to manage ambiguous applied research work.

Applied Scientist
2011-2016

A9.com Visual Search

- Early applied scientist on A9's visual search team; built recognition and tracking for A9 Flow, an augmented reality shopping application that identified multiple products in a scene and overlaid results in real time.
- Developed recognition, tracking, scene-text detection, and OCR libraries subsequently integrated into the Amazon mobile app and Amazon Fire Phone; supported uses including phone numbers, addresses, and business-card scanning.
- Built video content alignment tools for IMDb X-Ray that transferred existing annotations when studios redelivered video, reducing manual timestamp alignment work.
- Implemented classical computer vision algorithms in high-performance C++ and contributed inventions across visual search, text recognition, video analysis, and camera interfaces.

Research Scientist
2010-2011

Akiira Media Systems, Palo Alto

- Built a video content search engine at an early-stage computer vision startup.
- Developed automated news article summarization and story clustering for itwips.com, along with a video summarization API for iBlips.
- Applied singular value decomposition to hand-crafted multimedia features; contributed new clustering approaches and product interface design.

Research Associate
2009

Smith-Kettlewell Eye Research Institute, San Francisco

- Worked with James Coughlan on assistive technology for people with visual impairments; prototyped a Nokia N95 application that recognized fiducial targets for navigation and object identification.

Graduate Intern
2008-2009

Intel Corporation, Santa Clara

- Developed a prototype mobile augmented reality system for mobile internet devices.
- Improved large-scale image matching with keypoint indexing and geolocation filtering; built a database of geotagged Wikipedia articles and images.
- Published WikiReality at ICME 2009 and contributed the image-matching invention documented in a U.S. patent application.

Research Intern
2007-2008

NEC Labs America, Cupertino

- Researched automatic facial beauty prediction and enhancement using web images; developed learning methods operating on image appearance without manually annotated facial landmarks, published at ECCV 2010.
- Contributed face alignment software and a demographic recognition demonstration.

Engineering Intern
2006

SAIC Advanced Concepts Division

- Delivered a multi-object tracking library implementing Reid's multiple-hypothesis tracking algorithm.
- Extended the approach to multiple cameras and explored homography estimation from tracking data.

Graduate Student Researcher
2006-2009

Visual Computing Lab, UC Santa Cruz

- Conducted research with Hai Tao on viewpoint-invariant person recognition, appearance models, and crowd analysis.
- Collected and released the VIPeR benchmark: 632 pedestrian image pairs across camera viewpoints, with baseline evaluations published at PETS 2007 and ECCV 2008.
- Developed the Ensemble of Localized Features (ELF), using AdaBoost to select local color and texture features and learn a cross-camera similarity function; evaluated retrieval quality with cumulative matching characteristic (CMC) curves independently of camera geometry and timing.

Teaching Assistant
2006-2009

Computer Engineering, UC Santa Cruz

- Supported multiple terms of Microprocessor Design (CMPE 121), Stochastic Methods (CMPE 107), MATLAB (AMS 27L), Universal Access (CMPE 80A), and Technical Writing (CMPE 185).

Undergraduate Research Assistant
2004-2005

Visual Computing Lab, UC Santa Cruz

- Developed active camera tracking using mean-shift and pan-tilt-zoom control, multiple-hypothesis object tracking, and crowd-density estimation tools.

Art Director & CFO 2003-2005
Static Printing Inc.

- Co-founded a screen printing business and served as art director and CFO.
- The business later became Your Logo Here, LLC; no longer affiliated.

Education

Ph.D. Candidate (ABD) 2005-2010
Computer Engineering, UC Santa Cruz

- Advanced to candidacy in 2008; dissertation proposal: Privacy Preserving Pedestrian Recognition. Doctoral degree not completed.

B.S. 2001-2005
Computer Engineering, UC Santa Cruz

- B.S. in Computer Engineering; conducted undergraduate research in the Visual Computing Lab.

Professional Service

Workshop co-organizer. Multimodal Representation and Retrieval (MRR), SIGIR 2024 and ICCV 2025; served as lead organizer and advisor. Website.

Conference organizer and advisor. Amazon Computer Vision Conference (ACVC); lead organizer on two occasions, including the 2020 event, which moved to a virtual format attended by approximately 400 people; advisor in other years.

Area chair. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV).

Peer reviewer. CVPR, ICCV, ECCV, WACV, and NeurIPS.

Publications

[01] Vishwas Sathish, Viresh Ranjan, Xinliang Zhu, Arnab Dhua, **Douglas Gray**. **Eliciting Self-Verification in Multimodal Reasoning Agents with Reinforcement Learning**. *European Conference on Computer Vision (ECCV)*, pp. 384-401, 2026. DOI. Supplementary material.

[02] Sungyeon Kim, Xinliang Zhu, Xiaofan Lin, Muhammet Bastan, **Douglas Gray**, Suha Kwak. **GENIUS: A Generative Framework for Universal Multimodal Search**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 19659-19669, 2025.

[03] Xinliang Zhu, Sheng-Wei Huang, Han Ding, Jinyu Yang, Kelvin Chen, Tao Zhou, Tal Neiman, Ouye Xie, Son Tran, Benjamin Yao, **Douglas Gray**, Anuj Bindal, Arnab Dhua. **Bringing Multimodality to Amazon Visual Search System**. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 6390-6399, 2024. DOI.

[04] Zhizhang Hu, Shasha Li, Ming Du, Arnab Dhua, **Douglas Gray**. **De-noised Vision-language Fusion Guided by Visual Cues for E-commerce Product Search**. *CVPR Workshop on Multimodal Learning and Applications*, pp. 1986-1996, 2024.

[05] Zhizhang Hu, Shasha Li, Ming Du, Arnab Dhua, **Douglas Gray**. **Multimodal Learning with Online Text Cleaning for E-commerce Product Search**. *SIGIR Workshop on eCommerce*, 2024.

[06] Xinliang Zhu, Arnab Dhua, **Douglas Gray**, I Zeki Yalniz, Tan Yu, Mohamed Elhoseiny, Bryan Plummer. **Multimodal Representation and Retrieval [MRR 2024]**. *ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pp. 3047-3050, 2024. DOI.

[07] Qing Tian, Sampath Chanda, Amit Kumar K C, **Douglas Gray**. **Improving Apparel Detection with Category Grouping and Multi-grained Branches**. *Multimedia Tools and Applications*, 82(5), pp. 7383-7400, 2023. DOI.

[08] Ming Du, Arnau Ramisa, Amit Kumar K C, Sampath Chanda, Mengjiao Wang, Neelakandan Rajesh, Shasha Li, Yingchuan Hu, Tao Zhou, Nagashri Lakshminarayana, Son Tran, **Douglas Gray**. **Amazon Shop the Look: A Visual Search System for Fashion and Home**. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 2822-2830, 2022. DOI.

[09] Zhibo Yang, Muhammet Bastan, Xinliang Zhu, **Douglas Gray**, Dimitris Samaras. **Hierarchical Proxy-Based Loss for Deep Metric Learning**. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 1859-1868, 2022. Supplementary material.

[10] Ismet Zeki Yalniz, **Douglas Gray**, R Manmatha. **Efficient Exploration of Text Regions in Natural Scene Images Using Adaptive Image Sampling**. *2nd International Workshop on Robust Reading (IWRR), ECCV Workshops*, pp. 427-439, 2016. DOI.

[11] Jinjun Wang, Yihong Gong, **Douglas Gray**. **Female Facial Beauty Attribute Recognition and Editing**. *Human-Centered Social Media Analytics*, pp. 133-148, 2014. DOI.

[12] Ismet Zeki Yalniz, **Douglas Gray**, R Manmatha. **Adaptive Exploration of Text Regions in Natural Scene Image**. *International Conference on Document Analysis and Recognition (ICDAR)*, 2013. *Accepted; withdrawn before publication.*

[13] **Douglas Gray**, Kai Yu, Wei Xu, Yihong Gong. **Predicting Facial Beauty without Landmarks**. *European Conference on Computer Vision (ECCV)*, pp. 434-447, 2010. DOI.

[14] **Douglas Gray**, Igor Kozintsev, Yi Wu, Horst Haussecker. **WikiReality: Augmenting Reality with Community Driven Websites**. *IEEE International Conference on Multimedia and Expo (ICME)*, pp. 1290-1293, 2009.

[15] **Douglas Gray**, Hai Tao. **Viewpoint Invariant Pedestrian Recognition with an Ensemble of Localized Features**. *European Conference on Computer Vision (ECCV)*, pp. 262-275, 2008. DOI.

[16] **Douglas Gray**, Shane Brennan, Hai Tao. **Evaluating Appearance Models for Recognition, Reacquisition, and Tracking**. *IEEE International Workshop on Performance Evaluation of Tracking and Surveillance (PETS)*, 2007.

[17] Dan Kong, **Douglas Gray**, Hai Tao. **A Viewpoint Invariant Approach for Crowd Counting**. *International Conference on Pattern Recognition (ICPR)*, 3, pp. 1187-1190, 2006.

[18] Dan Kong, **Douglas Gray**, Hai Tao. **Counting Pedestrians in Crowds Using Viewpoint Invariant Training**. *British Machine Vision Conference (BMVC)*, 2005. DOI.

Patents

44 U.S. patents and 2 applications.

[P01] **Object classification for visual search**. Amit Kumar K C, Arnab Dhua, **Doug Gray**. US12430882B1; 2025-09-30.

[P02] **Multimodal search and refinement**. Xinliang Zhu, Arnab Dhua, Son D. Tran, **Douglas Ryan Gray**. US12210516B1; 2025-01-28.

[P03] **Media processing techniques for enhancing content**. **Douglas Ryan Gray**, Arnab Sanat Kumar Dhua, Xiaofan Lin, Zhijiang Mark Lu. US12106333B2; 2024-10-01.

[P04] **Media processing techniques for enhancing content**. **Douglas Ryan Gray**, Arnab Sanat Kumar Dhua, Xiaofan Lin, Zhijiang Mark Lu. US11699174B2; 2023-07-11.

[P05] **Augmented reality recommendations**. Xiaofan Lin, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Atul Kumar, Yu LOU. US11227326B2; 2022-01-18.

[P06] **Media processing techniques for enhancing content**. **Douglas Ryan Gray**, Arnab Sanat Kumar Dhua, Xiaofan Lin, Zhijiang Mark Lu. US10963924B1; 2021-03-30.

[P07] **Neural network-based image manipulation**. **Douglas Ryan Gray**, Alexander Li Honda, Edward Hsiao. US10956784B2; 2021-03-23.

[P08] **Object tracking in zoomed video**. Charles Benjamin Franklin Waggoner, Colin Jon Taylor, Jeffrey P. Bezos, **Douglas Ryan Gray**. US10664140B2; 2020-05-26.

[P09] **Visual search suggestions**. Edward Hsiao, **Douglas Ryan Gray**, Nityananda Jayadevaprakash, Xiaofan Lin, Mark Jay Nitzberg, Shruti Sheorey. US10540378B1; 2020-01-21.

[P10] **Video segmentation techniques**. Adam Carlson, **Douglas Ryan Gray**, Ashutosh Vishwas Kulkarni, Colin Jon Taylor. US10528821B2; 2020-01-07.

[P11] **Augmented reality recommendations**. Xiaofan Lin, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Atul Kumar, Yu LOU. US10506168B2; 2019-12-10.

[P12] **Expanded previously on segments**. Adam Carlson, Jeromey Russell Goetz, Ashutosh Vishwas Kulkarni, **Douglas Ryan Gray**, Danny Ryan Stephens, Colin Jon Taylor, Ismet Zeki Yalniz. US10469918B1; 2019-11-05.

[P13] **Crowdsourced audio normalization for presenting media content**. **Douglas Ryan Gray**, Colin Jon Taylor, Ming Du, Wei-Hong Chuang. US10466955B1; 2019-11-05.

[P14] **Neural network-based image manipulation**. **Douglas Ryan Gray**, Alexander Li Honda, Edward Hsiao. US10157332B1; 2018-12-18.

[P15] **Self-portrait enhancement techniques**. **Douglas Ryan Gray**, Colin Jon Taylor, Xiaofan Lin. US10121229B2; 2018-11-06.

[P16] **Attribute similarity-based search**. Ming Du, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Maya Kabkab, Aishwarya Natesh, Colin Jon Taylor. US10043109B1; 2018-08-07.

[P17] **Text entity recognition**. **Douglas Ryan Gray**, Xiaofan Lin, Arnab Sanat Kumar Dhua, Yu LOU. US10013624B2; 2018-07-03.

[P18] **Video content alignment**. Ismet Zeki Yalniz, Adam Carlson, **Douglas Ryan Gray**, Colin Jon Taylor. US9984728B2; 2018-05-29.

[P19] **Augmented reality Camera Lucida**. **Douglas Ryan Gray**. US9965895B1; 2018-05-08.

[P20] **Text recognition for search results**. Arnab Sunat Kumar Dhua, **Douglas Ryan Gray**, Xiaofan Lin, Yu LOU, Adam Wiggen Kraft, Sunil Ramesh. US9934526B1; 2018-04-03.

[P21] **Providing overlays based on text in a live camera view**. **Douglas R. Gray**, Arnab S. Dhua, Yu LOU, Sunil Ramesh. US9922431B2; 2018-03-20.

[P22] **Image processing techniques for self-captured images**. **Douglas Ryan Gray**. US9911237B1; 2018-03-06.

[P23] **Generating search strings and refinements from an image**. Edward Hsiao, **Douglas Ryan Gray**. US9875258B1; 2018-01-23.

[P24] **Automated highlighting of identified text**. Adam Wiggen Kraft, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Xiaofan Lin, Yu LOU, Sunil Ramesh, Colin Jon Taylor, David Creighton Mott. US9870633B2; 2018-01-16.

[P25] **Expanded previously on segments**. Adam Carlson, Jeromey Russell Goetz, Ashutosh Vishwas Kulkarni, **Douglas Ryan Gray**, Danny Ryan Stephens, Colin Jon Taylor, Ismet Zeki Yalniz. US9813784B1; 2017-11-07.

[P26] **Augmented reality recommendations**. Xiaofan Lin, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Atul Kumar, Yu LOU. US9646335B2; 2017-05-09.

[P27] **Object tracking in zoomed video**. Charles Benjamin Franklin Waggoner, Colin Jon Taylor, Jeffrey P. Bezos, **Douglas Ryan Gray**. US9626084B2; 2017-04-18.

[P28] **Automated highlighting of identified text**. Adam Wiggen Kraft, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Xiaofan Lin, Yu LOU, Sunil Ramesh, Colin Jon Taylor, David Creighton Mott. US9582913B1; 2017-02-28.

[P29] **Adaptive image sampling for text detection**. **Douglas Ryan Gray**, Arnab Sanat Kumar Dhua, Yu LOU, Ismet Zeki Yalniz. US9569679B1; 2017-02-14.

[P30] **Intelligent video navigation techniques**. **Douglas Ryan Gray**, Adam Carlson, Ashutosh Vishwas Kulkarni, Anna Makris, Colin Jon Taylor. US9564177B1; 2017-02-07.

[P31] **Intelligent video navigation techniques**. **Douglas Ryan Gray**, Adam Carlson, Ashutosh Vishwas Kulkarni, Anna Makris, Colin Jon Taylor. US9558784B1; 2017-01-31.

[P32] **Detection of cast members in video content**. Arnab Sanat Kumar Dhua, Gautam Bhargava, **Douglas Ryan Gray**, Sunil Ramesh, Colin Jon Taylor. US9449216B1; 2016-09-20.

[P33] **Collaborative text detection and recognition**. Xiaofan Lin, Adam Wiggen Kraft, Yu LOU, **Douglas Ryan Gray**, Colin Jon Taylor. US9436883B2; 2016-09-06.

[P34] **Video segmentation techniques**. Adam Carlson, **Douglas Ryan Gray**, Ashutosh Vishwas Kulkarni, Colin Jon Taylor. US9436876B1; 2016-09-06.

[P35] **Image-based character recognition**. Xiaofan Lin, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Yu LOU. US9390340B2; 2016-07-12.

[P36] **Verifying ownership of content**. **Douglas R. Gray**. US9317669B1; 2016-04-19.

[P37] **Automated recognition of text utilizing multiple images**. **Douglas Ryan Gray**, Colin Jay Taylor, Xiaofan Lin, Adam Wiggen Kraft, Yu LOU, Arnab Sanat Kumar Dhua. US9292739B1; 2016-03-22.

[P38] **Video content alignment**. Ismet Zeki Yalniz, Adam Carlson, **Douglas Ryan Gray**, Colin Jon Taylor. US9275682B1; 2016-03-01.

[P39] **Text entity recognition**. **Douglas Ryan Gray**, Xiaofan Lin, Arnab Sanat Kumar Dhua, Yu LOU. US9256795B1; 2016-02-09.

[P40] **Self-portrait enhancement techniques**. **Douglas Ryan Gray**, Colin Jon Taylor, Xiaofan Lin. US9247129B1; 2016-01-26.

[P41] **Augmented reality recommendations**. Xiaofan Lin, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Atul Kumar, Yu LOU. US9183583B2; 2015-11-10.

[P42] **Providing overlays based on text in a live camera view**. **Douglas R. Gray**, Arnab S. Dhua, Yu LOU, Sunil Ramesh. US9165406B1; 2015-10-20.

[P43] **Collaborative text detection and recognition**. Xiaofan Lin, Adam Wiggen Kraft, Yu LOU, **Douglas Ryan Gray**, Colin Jon Taylor. US9098888B1; 2015-08-04.

[P44] **Image-based character recognition**. Xiaofan Lin, Arnab Sanat Kumar Dhua, **Douglas Ryan Gray**, Yu LOU. US9043349B1; 2015-05-26.

[P45] **Detection of cast members in video content**. Arnab Sanat Kumar Dhua, Gautam Bhargava, **Douglas Ryan Gray**, Sunil Ramesh, Colin Jon Taylor. US20160358015A1 (application); 2016-12-08.

[P46] **Image matching for mobile augmented reality**. **Douglas R. Gray**, Yi Wu, Igor V. Kozintsev, Horst W. Haussecker, Maha El Choubassi. US20100309225A1 (application); 2010-12-09.